SPECIAL LIMITS

Deriving (1 – cos x)/x → 0

The conjugate turns a quotient that no rule handles directly into a product of two factors whose limits are already known.

01

NO RULE APPLIES — BEFORE

$$\frac{1-\cos x}{x}$$

↓ Multiply by the conjugate — top and bottom by 1 + cos x, then 1 – cos²x = sin²x

02

ONE FRACTION

$$\frac{\sin^2 x}{x(1+\cos x)}$$

↓ Rewrite as a product — split sin²x into sin x · sin x

03

PRODUCT OF KNOWN LIMITS

$$\frac{\sin x}{x}\cdot\frac{\sin x}{1+\cos x}$$

↓ Let x → 0 — the first factor tends to 1, the second to 0/2 = 0

04

LIMIT — AFTER

$$\lim_{x\to 0}\frac{1-\cos x}{x} = 1\cdot 0 = 0$$

▼ WHY NOT 1?